

Analytical Report

ACL 9th
Milestone 1

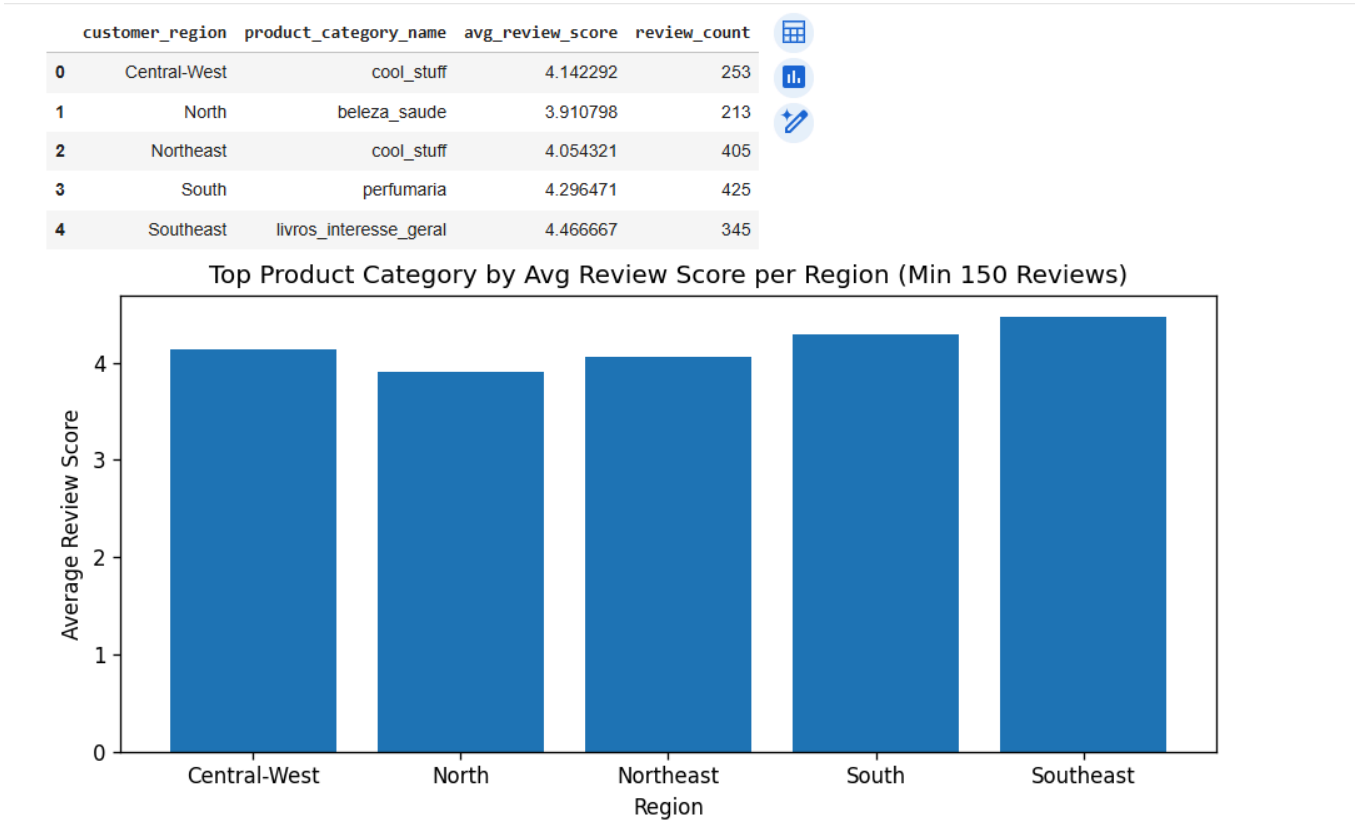
Team 1:
Yousef Awara
Mohab Essam
Youssef Medhat
Yousef Abdelmegeed

1. Product Category with Highest Average Review Score per Region

Based on the first chart, the following product categories achieved the highest average review scores in each region (with a minimum of 150 reviews):

- Central-West: cool_stuff (Avg: 4.14)
- North: beleza_saude (Avg: 3.91)
- Northeast: cool_stuff (Avg: 4.05)
- South: perfumaria (Avg: 4.29)
- Southeast: livros_interesse_geral (Avg: 4.47)

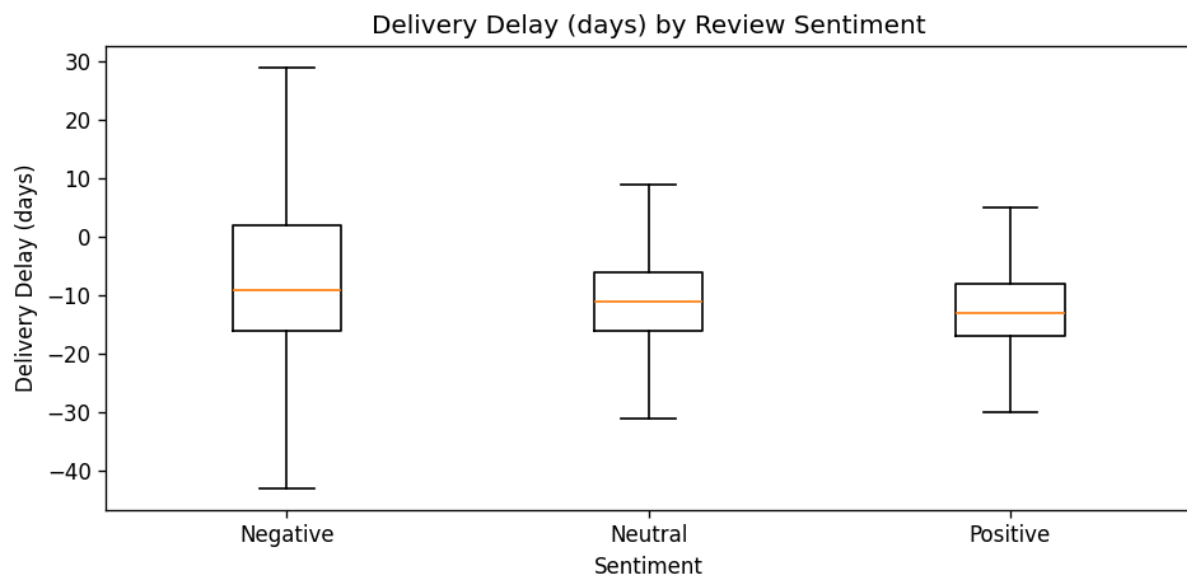
Among these, the Southeast region had the highest overall average review score (4.47) for the category 'livros_interesse_geral'. This suggests that consumers in the Southeast region are particularly satisfied with products related to books and general interests.



2. Relationship Between Delivery Delay and Review Sentiment

The boxplot shows a clear relationship between delivery delay and review sentiment. Negative reviews tend to have higher variability and longer delivery delays, with some extreme cases of late deliveries. Neutral and positive sentiments are associated with smaller delivery delays and less variation.

In particular, positive reviews generally correspond to earlier or on-time deliveries, while negative reviews are more frequent when delivery delays occur. This indicates a strong correlation between delivery performance and customer satisfaction.



3. Regions with the Highest Proportion of Negative Reviews

The third chart compares the proportion of negative reviews across regions. The Northeast region exhibits the highest share of negative reviews (approximately 19.8%), followed by the North (18.3%). The regions with the lowest proportion of negative feedback are the South (15.3%) and Southeast (15.6%).

This indicates that customers in the Northeast and North regions are more likely to express dissatisfaction compared to those in other areas, potentially due to issues such as delivery delays, product quality, or service inconsistencies.

	customer_region	negative_share
0	North	0.183491
1	Northeast	0.197903
2	Central-West	0.164129
3	Southeast	0.155806
4	South	0.153145

